

Project Title

MEDX Hospital Management System (MEDX HMS)

Document

Milestone 1 – Project Charter & Scope Document

Prepared By

Xante Graham

Project Type

Independent Portfolio Project

Date

July 2026

Table of Contents

1. Executive Summary	4
2. Background	4
3. Problem Statement	5
4. Project Overview	5
5. Vision Statement	6
6. Project Objectives	6
7. Deliverables	7
8. Project Scope	8
9. Scope Inclusions	8
Patient Features	8
Doctor Features	9
Administrator Features	9
General Features	9
10. Scope Exclusions	10
11. Stakeholders	11
Primary Stakeholders	11
Patients	11
Doctors	11
Hospital Administrators	11
Receptionists	11
Secondary Stakeholders	11
12. Assumptions	12
13. Constraints	12
Technical Constraints	12
Time Constraints	12
Resource Constraints	13
Knowledge Constraints	13
14. Success Criteria	13
15. Technology Stack	14
Frontend	14
Backend	14
Database	14

Development Tools	14
Design	15
Deployment (Planned)	15
16. Risks & Mitigation.....	15
17. Timeline	15

1. Executive Summary

The MEDX Hospital Management System (MEDX HMS) is a web-based healthcare management platform designed to improve hospital operations by centralizing essential administrative and clinical processes into one secure system. The platform enables patients, doctors, receptionists, nurses, and administrators to manage appointments, patient records, and day-to-day hospital activities more efficiently.

The project addresses common operational challenges faced by healthcare facilities, including fragmented patient information, inefficient appointment scheduling, communication gaps, and time-consuming administrative tasks. By providing a centralized platform, MEDX HMS aims to enhance operational efficiency, improve user experience, and support better decision-making.

As an independent portfolio project, MEDX HMS demonstrates the complete software development lifecycle; from project planning and requirements gathering to design, implementation, testing, and deployment, while showcasing technical project management principles and modern full-stack web development practices.

2. Background

Healthcare organizations depend on efficient coordination between patients, healthcare professionals, and administrative staff. In many healthcare settings, managing appointments, maintaining patient records, and coordinating daily operations can become complex, particularly when information is spread across multiple systems or manual processes.

Modern hospital management platforms help address these challenges by providing a centralized system where users can securely access information, manage appointments, and streamline communication. Such systems reduce administrative workload, improve data accuracy, and enhance the overall patient experience. MEDX Hospital Management System is designed as a prototype to demonstrate how an integrated digital platform can support hospital operations using modern web technologies and user-centered design principles.

3. Problem Statement

Hospitals manage a significant amount of information every day, including patient records, appointment schedules, staff availability, and administrative tasks. When these processes rely on disconnected systems or manual workflows, they can lead to scheduling conflicts, duplicate records, communication delays, and increased administrative effort. Healthcare providers require a centralized solution that simplifies these operations while ensuring information remains organized, accessible, and secure. The MEDX Hospital Management System addresses these challenges by providing an integrated web-based platform that improves communication, streamlines hospital workflows, and enhances the overall management of healthcare services.

4. Project Overview

MEDX Hospital Management System is a full-stack web application that provides a centralized platform for managing hospital operations. The system allows patients to register and schedule

appointments, enables doctors to manage consultations and view schedules, and provides administrators with tools to oversee users, appointments, and operational reports.

The project emphasizes usability, security, scalability, and maintainability while applying software engineering best practices throughout its development.

5. Vision Statement

To develop a secure, scalable, and user-friendly hospital management platform that improves healthcare service delivery by simplifying hospital operations, enhancing communication, and supporting efficient patient care through technology.

6. Project Objectives

The primary objective of the MEDX Hospital Management System (MEDX HMS) is to develop a secure, scalable, and user-friendly web application that improves the efficiency of hospital operations through centralized management of patients, appointments, staff, and administrative activities.

Specific objectives include:

- Design a centralized platform that enables hospitals to manage patients, appointments, and staff information efficiently.
- Allow patients to securely register, log in, and manage their appointments online.
- Enable doctors to access appointment schedules and patient information through a dedicated dashboard.

- Provide hospital administrators with tools to manage users, appointments, departments, and operational data.
- Improve communication between patients and healthcare providers through appointment notifications and status updates.
- Develop a responsive web application that functions effectively across desktop and mobile devices.
- Apply software engineering best practices, including requirements analysis, system modeling, documentation, testing, and version control.
- Demonstrate technical project management practices by planning, documenting, implementing, testing, and deploying the project using an organized development process.

7. Deliverables

The following deliverables will be produced throughout the development of the MEDX Hospital Management System.

Milestone	Deliverable
Milestone 1	Project Charter & Scope Document
Milestone 2	System Analysis & Design Document
Milestone 3	Fully Functional Web Application
Milestone 4	Testing Documentation, Deployment & User Guide

8. Project Scope

The MEDX Hospital Management System focuses on developing a prototype web application that demonstrates the core functions required to manage hospital operations digitally. The system will provide role-based access for patients, doctors, and administrators while maintaining secure authentication and centralized data management. The project will emphasize usability, maintainability, scalability, and modern software development practices. Although designed as a prototype, the system architecture will allow future expansion through additional modules.

9. Scope Inclusions

Patient Features

- User registration
- Secure login/logout
- Password reset
- View personal profile
- Edit personal information
- Book appointments
- Cancel appointments
- View appointment history
- Receive appointment status updates

Doctor Features

- Secure login
- View appointment schedule
- View patient information
- Update appointment status
- Record consultation notes (prototype)
- View daily dashboard

Administrator Features

- Manage patients
- Manage doctors
- Manage departments
- Manage appointments
- View hospital statistics
- Generate operational reports
- Manage user accounts

General Features

- Authentication
- Role-Based Access Control
- Responsive Design
- Search functionality
- Dashboard for each user role

- Secure database storage
- REST API
- Input validation
- Error handling

10. Scope Exclusions

To maintain a realistic project scope and timeline, the following features will **not** be included in

Version 1.0:

- Integration with real hospital systems
- Electronic prescribing
- Online payment processing
- Insurance claims processing
- Real-time ambulance tracking
- Video consultations
- AI-assisted diagnosis
- Machine Learning predictions
- Mobile application
- Multi-language support
- SMS notifications (Email notifications may be added later.)
- Integration with wearable health devices
- Cloud-based electronic medical records

These features are intentionally excluded to keep the project focused while leaving room for future enhancements.

11. Stakeholders

Primary Stakeholders

Patients

Individuals who use the platform to schedule appointments, access personal information, and manage healthcare interactions.

Doctors

Healthcare professionals responsible for reviewing patient information, managing consultations, and updating appointment statuses.

Hospital Administrators

Personnel responsible for managing users, appointments, departments, and system operations.

Receptionists

Staff members responsible for scheduling appointments and assisting patients.

Secondary Stakeholders

- Hospital Management
- IT Support Staff

- Software Developers
- Future System Administrators

12. Assumptions

The following assumptions are made during the development of the project:

- Users have internet access.
- Users possess basic computer literacy.
- Hospitals are willing to adopt digital management systems.
- Sample patient data will be used during development.
- Users will access the application through modern web browsers.
- The application is intended as a prototype rather than a production healthcare system.

13. Constraints

Technical Constraints

- The application will be developed using React, FastAPI, and PostgreSQL.
- Only open-source software and development tools will be used.
- Integration with external healthcare systems is outside the project scope.

Time Constraints

- The project will be completed before internship application deadlines.
- Development time is limited, requiring careful prioritization of features.

Resource Constraints

- The project is developed by a single developer.
- No external funding or paid software will be used.
- Development and testing will rely on personal computing resources.

Knowledge Constraints

- Advanced enterprise healthcare standards (such as HIPAA compliance or integrations with hospital information systems) are outside the scope of this educational portfolio project.
- Some technologies may be learned during development, so documentation and implementation may evolve as the project progresses.

14. Success Criteria

The project will be considered successful if:

- Patients can successfully register and log in.
- Patients can book and cancel appointments.
- Doctors can manage schedules.
- Administrators can manage users.
- Appointment conflicts are prevented.
- User information is stored securely.
- The application is fully responsive.

- The application is successfully deployed.
- Source code is available on GitHub with complete documentation.

15. Technology Stack

Frontend

- React
- HTML5
- CSS3
- JavaScript

Backend

- Python
- FastAPI

Database

- PostgreSQL

Development Tools

- Visual Studio Code
- Git
- GitHub

Design

- Figma
- Draw.io

Deployment (Planned)

- Frontend: Vercel
- Backend: Render
- Database: Neon PostgreSQL (or another hosted PostgreSQL service)

16. Risks & Mitigation

Risk	Mitigation
Learning a new technology takes longer than expected	Build the project incrementally and focus on core features first.
Bugs delay development	Test each feature as it is completed rather than waiting until the end.
Scope becomes too large	Prioritize Version 1.0 features and move additional ideas to future enhancements.
Database design needs revision	Review the ER diagram before implementation and update documentation if requirements change.

17. Timeline

Week	Activities
Week 1	Project planning, requirements gathering, and wireframes
Week 2	System design, UML diagrams, ERD, and API planning
Week 3	Backend development and database implementation
Week 4	Frontend development and integration
Week 5	Testing, deployment, documentation, portfolio updates, and demo video